Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/07/2015
PI Name:	Snell, Edward H. Ph.D.		
Project Title:	Growth Rate Dispersion as a Predictive Indicator for Biological Crystal Samples Where Quality Can be Improved with Microgravity Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	esnell@hwi.buffalo.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	716-898-8631
Organization Name:	Hauptman-Woodward Medical Research Institute		
PI Address 1:	Dept. of Structural Biology		
PI Address 2:	700 Ellicott Street		
PI Web Page:			
City:	Buffalo	**State:**	NY
Zip Code:	14203-1102	**Congressional District:**	26
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	01/24/2014	**End Date:**	01/23/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Karr, Laurel	**Contact Phone:**	(256) 544-7817
Contact Email:	LaurelKarr@nasa3.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2015: One PhD. student was associated with this proposal, Rick Roberts. He changed career tracks and was awarded a Masters Degree by thesis with his research entitled Structural and Bioinformatic analysis of Ethylmalonyl-CoA Decarboxylase. This also enabled a second student, Amanda Ruby, to become involved with part of the research. She also graduated this year with a masters Thesis entitled "A computational study of crystallization data: Application and potential prediction". The laboratory is currently seeking a new student to continue the research effort as part of their PhD or Masters research.		
COI Name (Institution):	Luft, Joseph (Hauptman Woodward Medical Research Institute)		
Grant/Contract No.:	NNX14AD80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light; thus, sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified in over 12 Nobel Prizes awarded to discoveries made from biological structures derived by X-ray crystallography. A limiting element to the process is the availability of high-quality, well-diffracting crystals. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. Biological macromolecules often take many days to grow crystals. Growth in reduced acceleration (commonly termed microgravity) on an orbiting spacecraft extends the physical quality of macromolecular crystals through a reduction in the mosaic spread (caused by slight misalignments of the molecules in the crystal lattice and used as a measure of long-range order) and an increase in crystal volume. Resolution (a measure of short-range order) is not directly affected by microgravity, but can still benefit with the correct experimental design to exploit the demonstrated improvements in long-range order. Microgravity growth can yield an improved quality crystal; however, not all samples improve from microgravity growth. If we could predict which samples could be improved by crystal growth in microgravity, then the true potential of this medium could be exploited in an efficient manner. It is our hypothesis that growth rate dispersion could be an indicator of crystals whose quality could be improved in microgravity. Growth rate dispersion is a phenomenon encountered in crystallization where seemingly identical crystals, produced from the same conditions, grow at different rates. We contend that large growth rate dispersion on the ground is indicative of a sample that should be improved by microgravity growth. In the case of inorganic crystals, a decrease in growth rate dispersion is correlated with an increase in physical crystal quality. The dispersion itself is thought to be caused by surface artifacts on the growing crystals that induce lattice imperfections. Gravity enables convective flows to develop in a crystallization solution. These flows are driven by density differences as molecules are depleted from the surrounding solution during nucleation and growth of the crystal. We plan to monitor growth rate dispersion of crystals grown on the ground and in microgravity, to determine if there is a correlation between the physical qualities of the resulting crystals with those measurements. We propose to use molecular biology techniques to alter the crystallization contacts and shift the growth rate dispersion properties of a single protein from low to high to test our predictive hypothesis. Finally we propose to extend the study to a selection of good and poorly diffracting crystals on the ground and confirm that those displaying high-growth rate dispersion on the ground are those that are improved on orbit and generate improved structural data when their quality is exploited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If our hypothesis proves to be correct, we will develop a generally applicable means to identify macromolecular crystals where quality will improve with microgravity growth. Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light, thus sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, which accounts for 90% of the known structural information. It is a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified by 12 Nobel Prizes awarded to discoveries made from X-ray crystallographic structures of biological macromolecules. Crystallography requires high-quality, well-diffracting crystals; crystallization is the most significant impediment to structure determination. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. In our high-throughput crystallization screening laboratory at the Hauptman-Woodward Medical Research Institute approximately 50% of proteins samples crystallize in at least one chemical condition tested and of those, approximately 50% go on to be deposited within the protein data bank – a depository for atomic resolution structural data. To restate this, some 25% of the macromolecules that enter the laboratory provide crystals that are of insufficient quality for structural determination, and for the 25% of macromolecules that do provide a structure; higher-quality crystals would enhance the structural detail to enable a better understanding of the molecular function. Microgravity crystallization has the potential to improve crystal quality and thereby provide novel structures for the 25% of macromolecules that produced crystals of insufficient quality for structure determination. In addition, it has the potential to improve the level of detail that can be observed in structures that have previously been resolved. It has become increasingly apparent that many of the failures to get structural data from crystals are not that they don't diffract, but that the diffraction is to very low resolution. A more ordered, physically higher quality crystal would increase the signal-to-noise of recorded data and extend this resolution to a point where structural information may be available, or potential pathways for improvement could be quantitatively evaluated.
	The task described involves producing multiple constructs of T4 lambda lysozyme that show different crystallization properties, in particular growth rate dispersion. That is the difference in individual growth rates of crystals grown in otherwise identical biochemical conditions. T4 Lysozyme variants are expressed in an IPTG-inducible system, the protocol for expression and purification is the same one that will be used for this work and is described later. Current expression levels are high, the wild type enzyme, WT, expresses at 17 mg/L with a cysteine mutant WT* (containing C45A and C97A used to improve expression) at 25 mg/L. Mutations currently expressed in our laboratory include S44E WT*, S44F WT*, E45A WT*, M6I, and I3P. These are expressed at levels of 10, 19, 9, 15, and 7 mg/L respectively, all suitable for crystallization trials. Another construct, R96A, failed to crystallize. Crystallization screening has been carried out on the native and mutants at identical concentrations. All except I3P produced initial crystals. In a Crossplate analysis (internal laboratory software) all the hits for the constructs and WT were compared. For the WT* construct there were 19 unique cocktails that gave crystals, five shared with S44F WT*, eight with E45A WT*, one with S44E WT*, one with M6I and four shared with both E45A WT* and S44E WT*. For the S44E WT* there were 11 unique cocktails and, excluding the one described previously, six shared with S44F WT*,

Task Progress:	two with E45A WT*, and two with both M6I and S44F WT*. For the S44F construct there were 114 unique cocktails and, beyond those already mentioned, four shared with E45A WT*. The E45A construct gave crystals in 16 unique cocktails out of 34 hits. M6I gave 21 crystal hits, 16 in unique conditions and finally I3P gave no crystal hits. There was no example where evidence of an initial crystal was seen in four or five of the different mutants. Crossplate is used as a quality control software to check consistency of sample handling and despite the stochastic nature of crystallization this result would be unusual unless the single point mutants significantly affect crystallization behavior. The difference in unique crystallization cocktails that yield crystals 11 versus 114 for the S44E WT* and S44F WT* shows that a single point mutation has considerable influence on the response to different crystallization cocktails and therefore crystal growth. A cursory examination of the crystallization cocktails producing crystals between the two suggests that S44E WT* prefers a higher pH range than S44F WT*, it is more sensitive to the both the cation and anion present, preferring a much narrower range of salts, and responds better to lower molecular weight PEGs. Crystallization screening was scaled up for all those samples that crystallized in the initial screening experiments. Five resulted in crystals that produced diffraction data at the Stanford Synchrotron Radiation Lightsource. A wild type structure of T4 lysozyme was determined to 2.12 Angstrom and the model deposited as in the Protein Data Bank (PDB) ID 4S0W. The wild type protein crystallized in Space Group P31. Structural data also resulted for the E45A, M6I, S44F WT* and WT* respectively. Resolution and Space Group were 2.6 Angstrom, P 31 2 1, 1.09 Angstrom, P31, and 1.50 Angstrom, C 2, respectively. The WT* is still being processed. Models for that data are currently being refined and will also be deposited in the PDB. All of the constructs used for structural data were set up in quartz glass capillaries (to be used for the flight and ground control experiments). Only three constructs have proved to reproducibly produce crystals in those capillaries, S44F, S44E, and WT*. Active fluid control is not possible in the current hardware configuration. To activate the experiment a freeze thaw process was developed within quartz glass capillaries. This allows samples to be prepared in advance, shipped to the loading facility/team, and preserved until the microscope (LMM) and crew time is available. Samples are preserved at -80 degrees C and a number of different filling and sealing processes were examined for the capillaries. The most successful sealing process was the use of cryogenic grease (which remains functional at cryogenic temperatures) at the end of each capillary followed by a 5 min epoxy seal. Other approaches were evaluated including light cured adhesive but the cryogenic grease and epoxy proved to be the most consistent. Growth rate has been determined for S44F crystals grown in capillaries. Data were collected for ten crystals. Growth rate was followed for a period of 120 hours and for the majority of the crystals, the rate slowed at approximately 40-60 hours. For a minority of crystals growth continued through the read time. Nucleation and growth is slowed under reduced acceleration and this time frame is likely to be longer on the flight. Growth rate has also been determined for the S44F WT* crystals which behaved differently to the S44F native case. Growth rate was followed for a period of 100 hours. In this case the minority of ten crystals studied (three) showed a levelling off of growth rate during the study. The remainder were still showing growth at the 100 hour point. It is important to note that the quartz capillaries used for flight and ground controls are significantly more expensive than the same size and capacity borosilicate capillaries. In the lead up to flight experiments, borosilicate capillaries have been used as a surrogate for the quartz capillaries. The influence of the material and surface properties has not been determined. In the S44F and S44F WT* studies quartz and borosilicate capillaries were used respectively. The S44F WT* conditions will also be studied in quartz although, while differences may occur, the growth rate difference is marked and is likely to remain so. The WT* crystals showed rapid growth, six crystals were studied and five showed growth leveling off after several hours with the remaining crystal slowing slow growth up to the 20 hour point of the experiment. This was also conducted in borosilicate capillaries. The three constructs described are being prepared for the flight experiments. Work continues to translate crystallization conditions for the other constructs into those that are reproducible within the capillary geometry.
Bibliography Type:	Description: (Last Updated: 12/14/2021)
Articles in Other Journals or Periodicals	Snell EH, Snell ME. "4S0W Wild type T4 lysozyme structure." Protein Data Bank: http://www.rcsb.org/pdb/explore/explore.do?structureId=4S0W (Initial release on: 4 February 2015); https://doi.org/10.2210/pdb4s0w/pdb (latest revision 28 December 2016) , Dec-2016
Dissertations and Theses	Ruby A. "A computational study of crystallization data: Application and potential prediction." Master of Science Thesis, State University of New York, University at Buffalo, May 2015. , May-2015
Dissertations and Theses	Roberts R. "Structural and Bioinformatic Analysis of Ethylmalonyl-coA Decarooxylase." Master's Thesis, State University of New York at Buffalo, July 2015. , Jul-2015